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Pediatrics Examination

Answer key and examiner record

Exam code: 143YVH2 | Date: 2026-10-07 | 50 questions | 50 minutes

Difficulty: 10 Easy / 15 Average / 25 Difficult
Seed: 20261006 | Actual draw seed: 20261006
Question format: Traditional single-best-answer only

Legacy Nelson 21st-edition questions. Difficulty labels are editorial estimates. This draw has not received a fresh clinical source review. Use for practice and mock assessments.

Topic coverage

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Answer grid

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Answers, explanations, and sources

1. The upper-to-lower body segment ratio is approximately what value at birth, reflecting the relatively large head and trunk?

Correct answer: A. About 1.7

The upper-to-lower body segment ratio is about 1.7 at birth, decreasing to about 1.3 at 3 years and about 1.0 after 7 years as the limbs lengthen.

B. About 1.0
A ratio of about 1.0 is reached after about 7 years, not at birth.

C. About 0.7
A ratio of about 0.7 is below the normal range and not seen at birth.

D. About 2.5
A ratio of about 2.5 overstates the newborn proportion.

NEL21-S02-Q102 | Growth, Development, and Behavior | Assessment of growth | Average
Review: Legacy content; no fresh source review

Source: Bamba V, Kelly A. Assessment of growth. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, eds. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 151-157.

2. A toilet-trained girl voids normally at regular intervals yet is constantly damp with urine day and night. Which anomaly best explains this?

Correct answer: B. An ectopic ureter inserting below the urinary sphincter

In girls an ectopic ureter can insert distal to the sphincter, into the urethra or vagina, producing continuous leakage despite normal-pattern voiding from the orthotopic system; it often drains an upper-pole duplex moiety.

A. Primary nocturnal enuresis without daytime symptoms
Nocturnal enuresis causes nighttime wetting, not constant dampness despite normal voiding.

C. An overactive bladder producing urgency incontinence
Overactive bladder causes urgency and intermittent leakage, not continuous wetness.

D. A small simple bladder wall diverticulum
A small diverticulum does not produce continuous incontinence.

NEL21-S23-Q047 | Urologic Disorders in Infants and Children | Ureterocele, Ectopic Ureter & Duplication | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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3. In girls with central precocious puberty, which underlying cause is most common?

Correct answer: C. Idiopathic activation of the hypothalamic-pituitary-gonadal axis

In girls, most central precocious puberty is idiopathic, whereas boys more often harbor an identifiable central nervous system lesion. This sex difference is why neuroimaging yields more often in boys with central precocity.

A. A hypothalamic hamartoma compressing the median eminence
Hamartomas cause central precocity but are far less common than idiopathic cases in girls.

B. A pineal germinoma secreting human chorionic gonadotropin
Germinomas are a rare cause and more relevant to boys.

D. Cranial irradiation injuring the hypothalamic region
Radiation is an uncommon identifiable cause in girls.

NEL21-S25-Q078 | The Endocrine System | Disorders of Pubertal Development | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 578, Disorders of Pubertal Development.

4. Which set names the four core principles of biomedical ethics?

Correct answer: C. Autonomy, beneficence, nonmaleficence, and justice

The four widely cited principles of biomedical ethics are respect for autonomy, beneficence, nonmaleficence, and justice.

A. Confidentiality, consent, capacity, and competence
These are important concepts but not the four core principles.

B. Veracity, fidelity, advocacy, and stewardship
These are professional virtues, not the four core principles.

D. Privacy, dignity, equity, and solidarity
These are values, but not the four-principle framework.

NEL21-S01-Q125 | The Field of Pediatrics | Pediatric Ethics | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 6, Ethics in Pediatric Care.

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5. A 7-year-old develops an acute eruption of numerous small, drop-like, scaly papules over the trunk about two weeks after a sore throat. Which condition and workup are most appropriate?

Correct answer: B. Guttate psoriasis; obtain streptococcal throat culture and titers

Guttate psoriasis classically follows streptococcal pharyngitis with crops of small drop-like scaly papules, so evaluation for streptococcal infection (throat culture and antistreptococcal titers) is appropriate.

A. Pityriasis rosea; reassurance with no testing
Pityriasis rosea begins with a herald patch and is not strep-associated.

C. Nummular eczema; perform patch testing
Nummular eczema forms coin-shaped exudative plaques, not a post-strep papular shower.

D. Secondary syphilis; obtain fungal cultures
Secondary syphilis is evaluated with serology, not fungal cultures.

NEL21-S30-Q031 | The Skin | Diseases of the Epidermis | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 676, Diseases of the Epidermis.

6. A child with acute gastroenteritis and mild-to-moderate dehydration is able to drink. Compared with intravenous therapy, which statement about oral rehydration is most accurate?

Correct answer: B. It is effective first-line therapy, even for many children with hyponatremic or hypernatremic dehydration

Oral rehydration is effective first-line therapy for most dehydration from gastroenteritis, including many children with mild to moderate hyponatremic or hypernatremic dehydration; intravenous therapy is reserved for severe dehydration or failure of oral intake.

A. It is contraindicated whenever the sodium is abnormal
Oral rehydration can manage many children with mild to moderate dysnatremic dehydration.

C. It becomes useful only after a trial of intravenous fluids has failed
Oral rehydration is first-line, not a salvage therapy after intravenous failure.

D. It is generally unable to replace the ongoing stool losses that continue during treatment
Oral rehydration can replace ongoing losses when intake keeps pace.

NEL21-S06-Q103 | Fluid, Electrolyte, and Acid-Base Disorders | Specific-disorder integration | Difficult
Review: Legacy content; no fresh source review

Source: Greenbaum LA. Deficit therapy. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 429-432.

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7. A 3-month-old former 28-week preterm infant requires inguinal hernia repair. Which postoperative complication most warrants apnea monitoring?

Correct answer: D. Postoperative apnea and bradycardia

Former preterm infants are at risk for postoperative apnea and bradycardia; risk is inversely related to postconceptual age and warrants extended monitoring (commonly to about 60 weeks postconceptual age).

A. Malignant hyperthermia
Malignant hyperthermia risk is unrelated to prematurity.

B. Postoperative nausea and vomiting
Nausea and vomiting is not the monitoring priority in this infant.

C. Emergence delirium
Emergence delirium is uncommon at this age and is not life-threatening.

NEL21-S07-Q113 | Pediatric Drug Therapy | Anesthesia & Perioperative Care | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Anesthesia and Perioperative Care. In: Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

8. A 6-month-old infant is found to have a perineal vascular lesion noted since shortly after birth, accompanied by minor bleeding. Beyond addressing the lesion itself, which of the following assessments is most important to perform?

Correct answer: B. A neurologic assessment for associated spinal dysraphism

Infantile hemangiomas are the most common benign vascular neoplasm of infancy, and most proliferate before resolving spontaneously. On identifying a perineal hemangioma, a neurologic assessment should be performed because of the association with spinal dysraphism.

A. Measurement of serum estradiol and luteinizing hormone levels
Estradiol and luteinizing hormone testing evaluates precocious puberty, which is not suggested by an isolated vascular lesion.

C. A forensic interview of the family for suspected abuse
A forensic interview is indicated when trauma or abuse is suspected, not for a congenital vascular neoplasm.

D. Vaginoscopy under anesthesia to exclude a foreign body
Vaginoscopy targets a suspected foreign body or unexplained internal bleeding, not a visible perineal hemangioma.

NEL21-S24-Q153 | Gynecologic Problems of Childhood | Causes of Prepubertal Vaginal Bleeding | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 565, Vaginal Bleeding in the Prepubertal Child.

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9. A neonate with Down syndrome has bilious vomiting and a double-bubble sign on abdominal radiograph. Which associated anomaly is most likely?

Correct answer: B. Duodenal atresia

Duodenal atresia is a well-recognized gastrointestinal anomaly in Down syndrome; bilious vomiting with a double-bubble sign on radiograph reflects proximal duodenal obstruction.

A. Pyloric stenosis
Pyloric stenosis causes nonbilious vomiting and is not the classic Down syndrome association here.

C. Biliary atresia
Biliary atresia presents with cholestatic jaundice, not a double-bubble obstruction.

D. Malrotation without volvulus
Malrotation can cause obstruction but does not produce the classic double-bubble of duodenal atresia.

NEL21-S09-Q072 | Human Genetics | Down syndrome | Average
Review: Legacy content; no fresh source review

Source: Cytogenetics. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 652-676.

10. Immediately after endotracheal intubation, which step is mandatory to confirm that the tube is in the trachea rather than the esophagus?

Correct answer: D. Detection of exhaled carbon dioxide by colorimetry or capnography

Confirmation of exhaled carbon dioxide using a colorimetric detector or capnography is mandatory to verify tracheal placement; esophageal intubation yields little or no sustained exhaled carbon dioxide. Relying on saturation can be misleading because preoxygenation delays desaturation even with a misplaced tube.

A. A single auscultation over the stomach only
Auscultation over the stomach alone is unreliable and does not confirm tracheal placement.

B. Observing the oxygen saturation alone for one minute
Preoxygenation can keep saturation high for minutes despite esophageal placement.

C. Measuring the cuff pressure
Cuff pressure does not indicate tube location.

NEL21-S08-Q073 | Emergency Medicine and Critical Care | Respiratory Failure | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 89, Respiratory Distress and Failure.

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11. There is a well-recognized bidirectional association between Tourette disorder and which condition?

Correct answer: C. Obsessive-compulsive disorder

Tourette disorder and OCD are bidirectionally associated, with 20-60% of TD patients meeting OCD criteria and 20-40% of OCD patients reporting tics.

A. Cystic fibrosis
Cystic fibrosis has no such neuropsychiatric association.

B. Type 1 diabetes mellitus
Type 1 diabetes is not linked bidirectionally to tics.

D. Chronic iron-deficiency anemia
Iron-deficiency anemia relates to breath-holding, not the TD-OCD link.

NEL21-S03-Q137 | Behavioral and Psychiatric Disorders | Motor disorders, tics & habits | Difficult
Review: Legacy content; no fresh source review

Source: Ryan CA, Walter HJ, DeMaso DR. Motor disorders and habits. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, eds. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 205-210.

12. Which set of effects is most consistently associated with exposure to media violence?

Correct answer: B. Increased aggression, desensitization, and reduced empathy in viewers

Media violence is associated with increased aggressive thoughts and behavior, desensitization, reduced empathy, and increased arousal, though a direct causal link to criminal behavior is not established.

A. A proven, direct causal link to later violent criminal behavior
A direct causal link to violent criminality is not established by the data.

C. A reliable reduction in aggressive thoughts and behaviors
Media violence increases, rather than reduces, aggression.

D. No measurable effect on arousal, empathy, or aggression at all
Measurable effects on aggression and empathy are well documented.

NEL21-S01-Q155 | The Field of Pediatrics | Violence & Children | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 14, Impact of Violence on Children.

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13. Under the National Institutes of Health criteria, the diagnosis of polycystic ovary syndrome requires which of the following combinations?

Correct answer: B. Oligoovulation or anovulation plus hyperandrogenism

The National Institutes of Health criteria require both oligoovulation or anovulation and clinical or biochemical hyperandrogenism, and do not incorporate ovarian ultrasound morphology. This contrasts with the Rotterdam criteria, which include polycystic ovaries as one of three possible elements.

A. Polycystic ovaries on ultrasonography plus hyperandrogenism
Pairing polycystic ovaries with hyperandrogenism reflects the Androgen Excess Society approach, not the National Institutes of Health criteria.

C. Polycystic ovaries on ultrasonography plus oligoovulation
This pairing omits the mandatory hyperandrogenism element required by the National Institutes of Health criteria.

D. Elevated luteinizing hormone-to-follicle-stimulating hormone ratio plus hyperandrogenism
An elevated luteinizing hormone-to-follicle-stimulating hormone ratio is explicitly not a diagnostic criterion.

NEL21-S24-Q038 | Gynecologic Problems of Childhood | PCOS Diagnostic Criteria & Presentation | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 567, Polycystic Ovary Syndrome and Hirsutism.

14. Colostrum, the first milk produced after birth, is especially rich in which protective component?

Correct answer: B. Secretory immunoglobulin A

Colostrum is high in protein and immunologic factors, especially secretory IgA, and is relatively lower in fat and lactose than mature milk; it provides early passive mucosal immune protection.

A. Milk sugar (lactose)
Colostrum is relatively low in lactose compared with mature milk.

C. Long-chain triglyceride fat
Colostrum has comparatively less fat than mature milk.

D. The protein casein
Casein is not the protein for which colostrum is notable; whey and IgA predominate.

NEL21-S05-Q200 | Nutrition | Feeding Infants, Children & Adolescents | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Feeding Healthy Infants, Children, and Adolescents. In: Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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15. In a child with gross hematuria and a normal coagulation profile, which microscopic finding best confirms an intrinsic glomerular cause?

Correct answer: A. Dysmorphic red cells accompanied by red blood cell casts

Dysmorphic red cells deformed during passage across the damaged filtration barrier, together with red cell casts, confirm a glomerular source. Uniform red cells without casts point to a urologic origin.

B. Numerous uniform red cells without casts
Uniform isomorphic red cells without casts indicate lower tract bleeding.

C. Abundant squamous epithelial cells
Squamous cells reflect contamination from the genital skin, not glomerular disease.

D. Calcium oxalate crystals with normal red cells
Oxalate crystals suggest stone disease rather than glomerular injury.

NEL21-S22-Q139 | Nephrology | Clinical evaluation of hematuria | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 536, Clinical Evaluation of the Child With Hematuria.

16. A 10-month-old has iron-deficiency anemia and intermittent blood-streaked stools after being given large amounts of cow's milk. Which mechanism most likely contributes to the iron deficiency?

Correct answer: C. Cow's-milk-protein-induced enteric blood loss

Cow's-milk-protein-induced colitis can cause occult or visible gastrointestinal blood loss that, combined with low dietary iron, produces iron deficiency in infants given large volumes of cow's milk.

A. Autoimmune hemolysis of red cells
Autoimmune hemolysis would cause jaundice and reticulocytosis, not bloody stools.

B. Intestinal hookworm infection with blood loss
Hookworm causes intestinal blood loss but depends on geographic exposure and is less likely here.

D. Splenic sequestration of red cells
Splenic sequestration causes splenomegaly, not enteric blood loss.

NEL21-S20-Q184 | Diseases of the Blood | Iron-deficiency & microcytic anemia | Difficult
Review: Legacy content; no fresh source review

Source: Iron-Deficiency Anemia. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 482.

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17. Köhler disease refers to avascular necrosis of which bone?

Correct answer: A. The tarsal navicular

Köhler disease is osteochondrosis (avascular necrosis) of the tarsal navicular, typically in young children, causing midfoot pain and a limp; it is usually self-limited. It is distinct from other foot osteochondroses.

B. The second metatarsal head
Avascular necrosis of the second metatarsal head is Freiberg infraction, not Köhler disease.

C. The calcaneal apophysis
Calcaneal apophysitis is Sever disease, an overuse traction problem, not navicular necrosis.

D. The base of the fifth metatarsal
The fifth metatarsal base is a common avulsion site, unrelated to navicular osteochondrosis.

NEL21-S31-Q058 | Bone and Joint Disorders | Foot & Toe Disorders | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 694, The Foot and Toes.

18. What is the principal physiologic role of carnitine in fatty acid metabolism?

Correct answer: A. Transporting long-chain fatty acids into mitochondria

Carnitine shuttles activated long-chain fatty acids across the inner mitochondrial membrane so they can undergo beta-oxidation. Defects in carnitine or its transferases impair energy production from fat during fasting.

B. Activating intestinal absorption of dietary fat
Carnitine does not drive intestinal fat absorption.

C. Synthesizing cholesterol from acetyl-CoA
Carnitine is not the agent of cholesterol synthesis.

D. Storing triglyceride within adipose tissue
Carnitine does not function as a triglyceride storage molecule.

NEL21-S10-Q037 | Metabolic Disorders | Fatty Acid Oxidation & Carnitine Defects | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 104, Defects in Metabolism of Lipids; p. 739-777.

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19. A newborn has a midline abdominal wall defect at the umbilical ring in which herniated viscera are covered by a membranous sac. Compared with gastroschisis, this lesion is most notable for which of the following?

Correct answer: D. A strong association with chromosomal and other major anomalies

An omphalocele, in which abdominal contents herniate into the base of the cord within a covering sac, is strongly associated with chromosomal abnormalities and syndromes such as trisomy 18 and Beckwith-Wiedemann, so affected infants warrant a careful search for other anomalies. The covered sac and central location distinguish it from gastroschisis.

A. Complete absence of any covering membrane over the bowel
Absence of a covering membrane characterizes gastroschisis, not omphalocele.

B. A location consistently to the right of the umbilical cord
A position to the right of the cord describes gastroschisis.

C. An almost exclusive occurrence as an isolated finding
Omphalocele is frequently syndromic rather than typically isolated.

NEL21-S11-Q092 | The Fetus and the Neonatal Infant | Neonatal GI Disorders & NEC | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 123, Digestive System Disorders.

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20. A 10-year-old from the south-central United States presents in July with five days of fever, headache, and myalgia after camping; he reports a tick bite. He has a faint maculopapular rash, mild splenomegaly, and altered mental status. Laboratory testing shows leukopenia, thrombocytopenia, elevated aminotransferases, and hyponatremia. Which infection best fits this presentation?

Correct answer: D. Human monocytic ehrlichiosis from Ehrlichia chaffeensis

Human monocytic ehrlichiosis is an undifferentiated febrile illness with fever, headache, and myalgia, often with rash in children, plus lymphadenopathy, hepatomegaly, or splenomegaly and altered mental status. The characteristic laboratory triad is leukopenia, thrombocytopenia, and elevated hepatic transaminases, with hyponatremia in most cases, following tick exposure in endemic regions. Doxycycline should be started empirically because therapy must not await confirmatory testing.

A. Acute Epstein-Barr virus infectious mononucleosis
Mononucleosis typically causes lymphocytosis with atypical lymphocytes rather than leukopenia, and lacks the tick-exposure link.

B. Disseminated meningococcemia with early septic shock
Meningococcemia produces a rapidly progressive purpuric rash and shock, not the leukopenia-thrombocytopenia-transaminitis triad after a tick bite.

C. Acute lymphoblastic leukemia presenting with cytopenias
Leukemia does not present acutely with fever following a documented tick bite and a self-limited febrile syndrome.

NEL21-S16-Q263 | Infectious Diseases | Mycoplasma, Chlamydia & Rickettsial Diseases | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 258, Ehrlichiosis and Anaplasmosis.

21. A child with progressive sensorineural loss has an enlarged vestibular aqueduct on CT and a positive perchlorate discharge test. Which gene is most likely involved?

Correct answer: D. SLC26A4 (pendrin)

This triad indicates Pendred syndrome, caused by SLC26A4 mutations affecting pendrin in the inner ear and thyroid.

A. GJB2 (connexin-26)
GJB2 causes nonsyndromic loss without this thyroid and aqueduct phenotype.

B. PAX3
PAX3 causes Waardenburg syndrome.

C. MYO7A in isolation
MYO7A in isolation relates to certain Usher and DFN forms, not this triad.

NEL21-S29-Q165 | The Ear | Hearing loss: genetics and syndromes | Difficult
Review: Legacy content; no fresh source review

Source: Haddad J Jr, Dodhia SN, Spitzer JB. Hearing loss. In: Kliegman RM, St. Geme JW, Blum NJ, et al., editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 3400-3411.

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22. A 15-year-old girl has 6 days of more than five bloody stools daily, fever, tachycardia, severe anemia, hypoalbuminemia, and leukocytosis; abdominal radiograph shows a dilated transverse colon. Which intervention is contraindicated in this setting?

Correct answer: B. Barium enema

This girl meets criteria for fulminant colitis with radiographic colonic dilation suggesting evolving toxic megacolon. Barium enema is contraindicated when toxic megacolon is suspected because instrumentation and luminal distention risk perforation; colonoscopy is similarly avoided. Management centers on intravenous steroids, fluid and electrolyte support, exclusion of infection, and surgical readiness if there is no response within several days.

A. Intravenous corticosteroids
Intravenous corticosteroids are a mainstay of therapy for fulminant colitis and are appropriate, not contraindicated, here.

C. Stool studies for enteric pathogens
Stool studies are appropriate and necessary to exclude superimposed enteric infection that can mimic or provoke a flare.

D. Intravenous fluid and electrolyte repletion
Intravenous fluids and electrolyte repletion are appropriate supportive measures for a severely ill child with fulminant colitis.

NEL21-S17-Q137 | The Digestive System | Inflammatory Bowel Disease | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

23. A former 25-week infant with bronchopulmonary dysplasia has persistent tachypnea, increased work of breathing during feeds, and poor weight gain despite adequate calories. Which of the following is the most appropriate management priority?

Correct answer: A. Optimize nutrition and energy intake for growth

Infants with bronchopulmonary dysplasia have high work of breathing and energy needs, so optimizing nutrition and caloric density supports lung growth and weight gain. Nutritional optimization is the priority.

B. Restrict total calories to lower metabolic demand
Calorie restriction worsens the existing growth failure.

C. Begin chronic daily oral corticosteroid therapy
Chronic systemic steroids carry serious adverse effects.

D. Stop supplemental oxygen to stimulate breathing
Removing needed oxygen worsens hypoxemia and pulmonary hypertension.

NEL21-S18-Q157 | The Respiratory System | Bronchopulmonary dysplasia | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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24. A neonate has severe symptomatic hypercalcemia, while an asymptomatic parent has lifelong mild hypercalcemia with low urine calcium. Which disorder links these findings?

Correct answer: D. A calcium-sensing receptor defect, severe when biallelic and benign when monoallelic

Calcium-sensing receptor mutations cause familial hypocalciuric hypercalcemia when heterozygous, a benign condition with low urinary calcium, and neonatal severe hyperparathyroidism when homozygous. The contrast between the affected neonate and the mildly affected parent reflects this gene-dosage effect.

A. Vitamin D intoxication in both the neonate and the parent
Shared receptor genetics, not intoxication, explains the family pattern.

B. Primary hyperparathyroidism from a parathyroid adenoma
An adenoma does not explain a benign, lifelong familial pattern with low urine calcium.

C. Dietary calcium excess in the breastfeeding pair
Diet does not account for the inherited low urinary calcium.

NEL21-S25-Q062 | The Endocrine System | Hyperparathyroidism & Hypercalcemia | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 591, Hyperparathyroidism.

25. Which is characteristic of the familial pattern of dyslexia?

Correct answer: A. Approximately 50% of children who have a parent with dyslexia will also have dyslexia

Dyslexia occurs in approximately 50% of children with an affected parent, 50% of siblings of dyslexic persons, and 50% of the parents of dyslexic individuals. Genome-wide association studies demonstrate that dyslexia is a complex trait caused by thousands of genetic variants, each contributing a small effect, rather than a single dominant gene.

B. Dyslexia is inherited in an autosomal dominant pattern linked to a single identifiable gene
Dyslexia is polygenic; no single gene has been identified as causal, and it does not follow a simple autosomal dominant pattern.

C. Risk in siblings of affected individuals is approximately 10%, reflecting low heritability
The sibling recurrence rate is approximately 50%, not 10%.

D. Familial occurrence is limited to paternal transmission and does not follow maternal lines
Dyslexia is transmitted through both maternal and paternal lines with equal familial rates.

NEL21-S04-Q179 | Learning and Developmental Disorders | Dyslexia | Easy
Review: Legacy content; no fresh source review

Source: Shaywitz SE, Shaywitz BA. Dyslexia. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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26. To best reduce coronary artery complications, the diagnosis and treatment of Kawasaki disease should ideally occur within which window?

Correct answer: A. Within 10 days, ideally within 7 days

Diagnosis and treatment within 10 days, and ideally within 7 days, of fever onset best reduce coronary artery complications.

B. Within 30 days of the initial fever onset
A 30-day window is too late to optimally prevent coronary damage.

C. After the fever has fully resolved
Waiting for defervescence delays needed treatment.

D. Once coronary aneurysms have appeared
Waiting for aneurysms to appear defeats the purpose of early treatment.

NEL21-S15-Q197 | Rheumatic Diseases of Childhood | Kawasaki disease | Difficult
Review: Legacy content; no fresh source review

Source: Son MBF, Newburger JW. Kawasaki Disease. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 1311-1316.

27. Calcium EDTA chelation requires monitoring for which characteristic toxicity?

Correct answer: D. Nephrotoxicity with proteinuria and rising creatinine

Calcium EDTA can cause renal tubular injury, so urinalysis and renal function are monitored during treatment with adequate hydration. The other toxicities are not characteristic of calcium EDTA.

A. Progressive irreversible sensorineural hearing loss
Sensorineural hearing loss is not the characteristic EDTA toxicity

B. Acute hemorrhagic cystitis with passage of clots
Hemorrhagic cystitis is associated with other agents, not EDTA

C. Immediate anaphylaxis upon the very first dose
First-dose anaphylaxis is not the typical EDTA reaction

NEL21-S33-Q189 | Environmental Health | Lead poisoning | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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28. Which abnormality underlies the most common T-cell immunodeficiency disorder?

Correct answer: C. A chromosome 22q11.2 deletion

Chromosome 22q11.2 deletion syndrome is the most common T-cell disorder, occurring in about 1 in 3,000 births, and disrupts development of the third and fourth pharyngeal pouches with thymic and parathyroid hypoplasia.

A. Mutation of the AIRE gene
AIRE mutation causes the autoimmune polyendocrinopathy-candidiasis syndrome, which is uncommon.

B. Mutation of the WAS gene
WAS mutation causes Wiskott-Aldrich syndrome.

D. A defect in ZAP-70 signaling
ZAP-70 defects cause a selective CD8 deficiency and are rare.

NEL21-S13-Q011 | Immunology | Cellular defects & 22q11.2 deletion | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 151, Primary Defects of Cellular Immunity; p. 1112-1115.

29. A family requests therapy to change their adolescent's same-sex orientation. Which response reflects the position of major medical organizations?

Correct answer: A. A gay or lesbian orientation is not an illness and need not be changed

Leading organizations, including the American Academy of Pediatrics, conclude that a gay, lesbian, or bisexual orientation is part of the spectrum of human sexuality, is not a mental illness, and that change efforts are neither possible nor warranted. Clinicians should provide supportive, nonjudgmental care.

B. Brief behavioral therapy reliably converts orientation to heterosexual when started early enough
No therapy reliably or safely changes sexual orientation.

C. Orientation-change counseling is appropriate whenever a family strongly prefers that approach
Orientation-change efforts are not recommended regardless of preference.

D. A same-sex orientation is a psychiatric disorder that warrants corrective treatment
Same-sex orientation is not classified as a psychiatric disorder.

NEL21-S12-Q019 | Adolescent Medicine | Gay, Lesbian & Bisexual Youth | Easy
Review: Legacy content; no fresh source review

Source: Adelson SL, Schuster MA. Gay, lesbian, and bisexual adolescents. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 1024-1026.

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30. Which mediator drives the iron sequestration of anemia of chronic disease by promoting internalization of ferroportin?

Correct answer: C. Hepcidin

Inflammatory cytokines, especially interleukin-6, raise hepcidin, which binds and internalizes ferroportin on enterocytes and macrophages, reducing iron absorption and release and producing an iron-restricted anemia.

A. Erythropoietin
Erythropoietin stimulates red cell production rather than restricting iron.

B. Ferritin
Ferritin is an iron-storage protein, not the regulator.

D. Transferrin
Transferrin transports iron in plasma.

NEL21-S20-Q042 | Diseases of the Blood | Anemia of chronic disease & renal disease | Average
Review: Legacy content; no fresh source review

Source: Anemia of Chronic Disease and Renal Disease. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 478.

31. In a neonate or young infant aged 0-3 months, what minimum rectal temperature defines fever without a focus?

Correct answer: C. 38.0C (100.4F)

For infants aged 0-3 months, fever without a focus refers to a rectal temperature of 38.0C (100.4F) or greater without other presenting signs or symptoms. This threshold is important because serious bacterial infection occurs in roughly 7-13% of febrile neonates and young infants. Hypothermia below 36.0C is also a recognized sign of systemic illness in this age-group.

A. 37.5C (99.5F)
This value is below the accepted threshold and would overclassify normal temperatures as fever.

B. 38.5C (101.3F)
This value is higher than the accepted cutoff and would miss qualifying infants.

D. 39.0C (102.2F)
This is the high-fever threshold used in older children, not the neonatal definition.

NEL21-S16-Q017 | Infectious Diseases | Fever Syndromes & Antibacterial Principles | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 202, Fever Without a Focus in the Neonate and Young Infant.

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32. According to symptom-based pediatric criteria, infant colic requires recurrent crying or fussing without obvious cause occurring in an infant of which age range?

Correct answer: D. Younger than 5 months

Diagnostic criteria for infant colic require an infant younger than 5 months when symptoms start and stop, with recurrent prolonged crying, fussing, or irritability that occurs without obvious cause and cannot be prevented or resolved by caregivers, and no failure to thrive, fever, or illness. It is a self-limited condition managed with reassurance and support.

A. Younger than 2 months
Restricting onset to under 2 months is narrower than the criteria, which extend the qualifying age further.

B. Between 6 and 12 months
An age of 6 to 12 months is beyond the window in which colic begins and resolves.

C. Between 12 and 18 months
An age of 12 to 18 months is well outside the defined period for infant colic.

NEL21-S17-Q172 | The Digestive System | Chronic Diarrhea & Functional GI Disorders | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

33. Which medication is commonly used to lower cerebrospinal fluid production in idiopathic intracranial hypertension?

Correct answer: D. Acetazolamide

Acetazolamide, a carbonic anhydrase inhibitor, reduces cerebrospinal fluid production and is a mainstay of medical therapy for idiopathic intracranial hypertension, alongside weight management.

A. Oral prednisone
Corticosteroids are not the mainstay and may precipitate the condition on withdrawal.

B. Indomethacin
Indomethacin is used for certain other headache syndromes, not as the standard agent here.

C. Sumatriptan
Sumatriptan is a migraine abortive and does not lower cerebrospinal fluid pressure.

NEL21-S26-Q042 | The Nervous System | Headache, Migraine & Idiopathic Intracranial Hypertension | Easy
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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34. A 7-year-old girl has mild proximal weakness and a creatine kinase value of a few thousand units; her brother has a confirmed dystrophinopathy; which mechanism best explains her symptoms?

Correct answer: A. Skewed X-inactivation per the Lyon hypothesis

About 8% of carrier females are manifesting carriers with weakness, explained by skewed X-inactivation under the Lyon hypothesis in which the normal X chromosome is inactivated and the deleted one is active. Such carriers are typically milder than affected males.

B. A new autosomal dominant point mutation
An autosomal dominant point mutation does not explain symptoms in an X-linked dystrophinopathy carrier.

C. Mitochondrial heteroplasmy in muscle
Mitochondrial heteroplasmy is unrelated to the nuclear X-linked dystrophin defect.

D. Homozygous dystrophin gene deletion
A homozygous dystrophin deletion in a girl would produce a full Duchenne picture, not the mild manifesting-carrier state described.

NEL21-S27-Q142 | Neuromuscular Disorders | Duchenne and Becker muscular dystrophy | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 627, Muscular Dystrophies; p. 3280-3294.

35. A survivor of Hodgkin lymphoma treated with chest radiation and an anthracycline asks about long-term risks. Which combination is most important to monitor?

Correct answer: B. Breast cancer and cardiac dysfunction

Mantle (chest) radiation raises the risk of secondary breast cancer and cardiac disease, and anthracyclines add cardiomyopathy risk, so breast and cardiac surveillance are priorities.

A. Hearing loss and chronic nephropathy
Hearing loss and nephropathy relate to platinum agents, not this regimen.

C. Hemorrhagic cystitis and male infertility
Cystitis relates to oxazaphosphorines, which were not used here.

D. Pulmonary fibrosis and posterior cataracts
Pulmonary fibrosis and cataracts are not the dominant risks of this combination.

NEL21-S21-Q174 | Cancer and Benign Tumors | Lymphoma (Hodgkin & Non-Hodgkin) | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 523, Lymphoma.

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36. A variant at the boundary between an intron and an exon disrupts normal splicing. Which consequence is expected?

Correct answer: C. Exon skipping or retention of intronic sequence

Splice-site variants disrupt the signals that direct intron removal, leading to skipping of an exon or retention of intronic sequence in the mature messenger RNA, which alters the protein.

A. Addition of a longer poly-A tail
Poly-A tail length is set during processing and is not the effect of a splice-site variant.

B. Substitution of a single amino acid only
A single amino acid substitution describes a missense variant, not a splicing defect.

D. Increased rate of gene transcription
Increased transcription relates to promoter or regulatory changes, not splicing.

NEL21-S09-Q175 | Human Genetics | Molecular genetics and the human genome | Difficult
Review: Legacy content; no fresh source review

Source: The human genome. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 635-640.

37. Which statement about physical growth in middle childhood is MOST accurate?

Correct answer: B. Growth is discontinuous, occurring in several irregular spurts each year

Growth in middle childhood occurs discontinuously in 3-6 irregularly timed spurts per year, with head circumference increasing only about 2 cm across the entire period.

A. Growth is perfectly steady and continuous day to day
Growth is not perfectly steady; it occurs in irregular spurts.

C. Head circumference increases by about 10 cm over the period
Head circumference increases only about 2 cm across the period, not 10 cm.

D. Linear growth ceases entirely until puberty
Linear growth continues at about 6-7 cm per year during middle childhood.

NEL21-S02-Q150 | Growth, Development, and Behavior | Middle childhood | Difficult
Review: Legacy content; no fresh source review

Source: Finkelstein LH, Feigelman S. Middle childhood. In: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, eds. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 146-151.

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38. A teenager who wears soft contact lenses develops itching and large papillae on the upper tarsal conjunctiva. Which condition is most likely?

Correct answer: D. Giant papillary conjunctivitis

Giant papillary conjunctivitis is associated with contact lens wear and other chronic ocular surface irritants, producing large upper tarsal papillae and itch. Management includes reducing lens wear and improving lens hygiene.

A. Vernal keratoconjunctivitis
Vernal keratoconjunctivitis is seasonal and typically affects young boys without a lens trigger.

B. Seasonal allergic conjunctivitis
Seasonal allergic conjunctivitis lacks the giant papillae and lens association.

C. Bacterial conjunctivitis
Bacterial conjunctivitis causes purulent discharge rather than itch and papillae.

NEL21-S14-Q087 | Allergic Disorders | Ocular Allergies | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 172, Ocular Allergies; p. 1219-1222.

39. A child in ventricular fibrillation has received one shock at 2 J/kg followed by 2 minutes of CPR, and the rhythm remains ventricular fibrillation. After delivering a second shock, what is the recommended next action?

Correct answer: C. Immediately resume chest compressions and give epinephrine 0.01 mg/kg

In persistent VF/pulseless VT, CPR is resumed immediately after each shock, and epinephrine 0.01 mg/kg IV/IO is given (repeated every 3-5 minutes) once the rhythm is shock-refractory, followed by an antiarrhythmic such as amiodarone. Interrupting compressions for prolonged pulse or rhythm checks worsens outcomes.

A. Check a pulse for a full 60 seconds before resuming compressions
A prolonged pulse check interrupts perfusion; compressions should resume right after the shock.

B. Administer sodium bicarbonate before resuming compressions
Routine bicarbonate is not indicated and should not precede compressions.

D. Withhold compressions and obtain a 12-lead electrocardiogram
Obtaining a 12-lead ECG during arrest dangerously delays compressions.

NEL21-S08-Q122 | Emergency Medicine and Critical Care | Resuscitation | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 81, Pediatric Emergencies and Resuscitation.

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40. A sinus venosus atrial septal defect is most often associated with which additional anomaly?

Correct answer: C. Partial anomalous pulmonary venous connection of a right vein

The superior sinus venosus defect lies near the superior vena cava-right atrium junction and is frequently accompanied by anomalous drainage of the right upper pulmonary vein, adding to the right heart load.

A. A discrete subaortic membrane causing obstruction
A subaortic membrane is unrelated to sinus venosus defects.

B. Apical displacement of the tricuspid leaflets
Tricuspid displacement defines Ebstein anomaly, a separate lesion.

D. Supravalvar narrowing of the ascending aorta
Supravalvar aortic stenosis is a feature of Williams syndrome.

NEL21-S19-Q195 | The Cardiovascular System | Acyanotic CHD: Left-to-Right Shunts | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 453, Acyanotic Congenital Heart Disease: Left-to-Right Shunt Lesions.

41. A 15-year-old girl with mild intellectual disability (ID) who previously received services under an individualized education program (IEP) is transitioning to a community college vocational program. Her parents are concerned about her future independence. According to outcomes data, what is the approximate employment rate for adults with ID?

Correct answer: A. Only about 15% of adults with ID are estimated to have jobs

Despite employment being a critical element of life adaptation for persons with ID, only approximately 15% are estimated to have jobs, with significant gaps in pay and compensation compared to workers without disability. This underscores the importance of early planning, IEPs, vocational training, and expansion of employment opportunities beginning in adolescence.

B. Employment rates for adults with ID approximate those of the general population
Employment rates for adults with ID are far below those of the general population; the chapter highlights significant gaps in pay and employment.

C. About 75% of adults with mild ID are employed in supported employment settings
About 75% is not supported by the text; the stated estimate is approximately 15% with jobs overall, reflecting major underemployment.

D. Approximately 50% of adults with ID are employed in competitive positions
An employment rate of approximately 50% substantially overstates the reported estimate of about 15% for adults with ID.

NEL21-S04-Q126 | Learning and Developmental Disorders | Developmental Delay & Intellectual Disability | Difficult
Review: Legacy content; no fresh source review

Source: Shapiro BK, O'Neill ME. Developmental delay and intellectual disability. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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42. In mass psychogenic illness, symptoms most often spread through which mechanism?

Correct answer: D. Seeing or hearing others who are affected

Symptoms propagate by visual and auditory cues as group members witness others becoming ill, consistent with a behavioral rather than biologic mechanism. The other routes imply an infectious or toxic source.

A. Direct bloodborne transmission passing between individuals
There is no bloodborne agent in psychogenic spread

B. Ingestion of a single common contaminated food
A shared contaminated food implies a toxic or infectious cause

C. The bite of a single shared arthropod vector
An arthropod vector implies a biologic, not psychogenic, cause

NEL21-S33-Q049 | Environmental Health | Mass psychogenic illness | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

43. An 8-year-old boy has intense ocular itching, photophobia, ropy discharge, and giant cobblestone papillae on the upper tarsal conjunctiva that worsen each spring. Which is the most likely diagnosis?

Correct answer: C. Vernal keratoconjunctivitis

Vernal keratoconjunctivitis is a chronic, often seasonal allergic disease of boys characterized by giant cobblestone tarsal papillae, intense itching, and a risk of sight-threatening corneal involvement. It is more severe than ordinary allergic conjunctivitis.

A. Seasonal allergic conjunctivitis
Seasonal allergic conjunctivitis causes itch and tearing but not giant cobblestone papillae or shield ulcers.

B. Giant papillary conjunctivitis from contact lenses
Giant papillary conjunctivitis is linked to contact lens wear, not seasonal flares in a child.

D. Bacterial conjunctivitis
Bacterial conjunctivitis produces purulent discharge without intense itch or cobblestone papillae.

NEL21-S14-Q016 | Allergic Disorders | Ocular Allergies | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 172, Ocular Allergies; p. 1219-1222.

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44. An injury is classified as ASIA Impairment Scale grade A when which condition is met?

Correct answer: A. No sensory or motor function is preserved in sacral segments S4-S5

ASIA Impairment Scale grade A is a complete injury, defined by the absence of any sensory or motor function in the sacral segments S4-S5.

B. Some motor function is preserved more than three levels below the lesion
Preserved motor function more than three levels below the lesion indicates an incomplete, motor-incomplete injury.

C. Light touch is preserved at S4-S5 but motor function is absent below
Preserved sacral light touch with absent motor function indicates a sensory-incomplete grade B.

D. At least half of key muscles below the lesion are graded 3 or better
Half or more of key muscles graded 3 or better below the lesion describes grade D.

NEL21-S32-Q029 | Rehabilitation Medicine | Spinal Cord Injury & Autonomic Dysreflexia | Easy
Review: Legacy content; no fresh source review

Source: Bryant PR, Jaffe A. Spinal cord injury and autonomic dysreflexia management. In: Kliegman RM, St. Geme JW, Blum NJ, et al., editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 3755-3760.

Source: Kirshblum SC, Burns SP, Biering-Sorensen F, et al. International standards for neurological classification of spinal cord injury (revised 2011). J Spinal Cord Med. 2011;34(6):535-546.

45. Renal clearance of gentamicin is lowest, requiring the longest dosing interval, in which group?

Correct answer: C. Neonates in the first weeks of life

Glomerular filtration is only 2-4 mL/min/1.73 m2 at birth and approaches adult values by 10-12 months, so neonates need the longest gentamicin interval (often every 18-24 hours); febrile young children may even need shorter intervals.

A. Children aged two to five years
Young children often have glomerular filtration at or above adult values.

B. Adolescents aged twelve to eighteen years
Adolescent renal function is mature, at adult levels.

D. School-age children aged six to ten years
School-age renal function approximates the adult pattern.

NEL21-S07-Q067 | Pediatric Drug Therapy | Developmental Pharmacokinetics | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Principles of Drug Therapy. In: Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

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46. Asymptomatic bacteriuria in an otherwise healthy child is best managed in which way?

Correct answer: A. Generally without antibiotic treatment

Asymptomatic bacteriuria in a healthy child generally should not be treated, since treatment selects for resistant uropathogens without preventing symptomatic infection or renal damage.

B. With a prolonged antibiotic course
Treating asymptomatic bacteriuria promotes resistant organisms without benefit.

C. With prompt surgical evaluation
Surgery is not indicated for asymptomatic bacteriuria.

D. With chronic catheter drainage
Catheterization is inappropriate and would introduce infection.

NEL21-S23-Q193 | Urologic Disorders in Infants and Children | Urinary Tract Infection | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020.

47. Bilateral itching, watery discharge, and conjunctival chemosis that recurs seasonally most characterize which type of conjunctivitis?

Correct answer: D. Allergic conjunctivitis

Allergic conjunctivitis features intense bilateral itching, watery discharge, and chemosis, often seasonal; cold compresses, antihistamines, and mast cell stabilizers provide relief.

A. Gonococcal conjunctivitis
Gonococcal disease is a hyperacute purulent neonatal conjunctivitis.

B. Bacterial purulent conjunctivitis
Bacterial conjunctivitis gives mucopurulent discharge, not itching and chemosis.

C. Chlamydial conjunctivitis
Chlamydial conjunctivitis is subacute, not itchy and seasonal.

NEL21-S28-Q148 | Disorders of the Eye | Conjunctiva & ophthalmia neonatorum | Easy
Review: Legacy content; no fresh source review

Source: Olitsky SE, Marsh JD. Disorders of the conjunctiva. In: Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. p. 3364-3368.

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48. Which is the treatment of choice for isolated moderate-to-severe valvar pulmonary stenosis?

Correct answer: B. Percutaneous balloon valvuloplasty

Balloon valvuloplasty is the first-line treatment for typical doming valvar pulmonary stenosis with a significant gradient, giving excellent long-term relief of obstruction.

A. Long-term oral beta-blocker therapy
Beta blockade does not relieve fixed valvar pulmonary obstruction.

C. Continuous prostaglandin E1 infusion
Prostaglandin is used for ductal-dependent lesions, not isolated valvar stenosis.

D. Surgical pulmonary artery banding
Banding adds obstruction and is not used for pulmonary stenosis.

NEL21-S19-Q094 | The Cardiovascular System | Acyanotic CHD: Obstructive Lesions | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 454, Acyanotic Congenital Heart Disease: Obstructive Lesions.

49. Which feature most strongly favors spontaneous intestinal perforation over necrotizing enterocolitis in a preterm infant?

Correct answer: B. An isolated perforation early in life without preceding pneumatosis

Spontaneous intestinal perforation is typically a focal, isolated perforation occurring in the first days of life, often associated with early indomethacin or corticosteroid exposure, and characteristically lacks the antecedent pneumatosis of necrotizing enterocolitis. This distinction guides management and prognosis.

A. Extensive pneumatosis intestinalis preceding the perforation
Extensive pneumatosis is a hallmark of necrotizing enterocolitis rather than spontaneous perforation.

C. Onset only after full enteral feeds with portal venous gas
Onset after established feeds with portal venous gas is typical of necrotizing enterocolitis.

D. Bloody stools accompanied by widespread intramural gas
Bloody stools with diffuse intramural gas describe necrotizing enterocolitis.

NEL21-S11-Q144 | The Fetus and the Neonatal Infant | Neonatal GI Disorders & NEC | Difficult
Review: Legacy content; no fresh source review

Source: Kliegman RM, St Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 123, Digestive System Disorders.

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50. Epipodophyllotoxins such as etoposide are notably associated with which delayed complication?

Correct answer: D. Secondary acute myeloid leukemia

Topoisomerase II inhibitors such as etoposide are linked to therapy-related acute myeloid leukemia, frequently carrying KMT2A (11q23) rearrangements and a relatively short latency.

A. Delayed hemorrhagic cystitis
Cystitis is an oxazaphosphorine effect.

B. Late dilated cardiomyopathy
Cardiomyopathy reflects anthracyclines.

C. Progressive pulmonary fibrosis
Pulmonary fibrosis points to bleomycin or busulfan.

NEL21-S21-Q129 | Cancer and Benign Tumors | Principles of Cancer Treatment & Oncologic Emergencies | Average
Review: Legacy content; no fresh source review

Source: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia: Elsevier; 2020. Chapter 521, Principles of Cancer Treatment.

Bank fingerprint: b871e29aef79951fe25215c808e10bb69fae143394f695b965c8960388737fe2

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